Clinical Trials

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Review of clinical trials methods

- Type of trial: parallel, multi-arm, cross-over, etc
- Unit of allocation: individual, cluster
- Framework: superiority, non-inferiority
- Allocation: Non-random, Quasi-random, Random
- Controlled or not
- Phase: Only for drug trials

Sample size calculation

Preample

- Estimated number of participants needed to achieve study objectives
- Describe how it was determined
- Explicitly indicate clinical and statistical assumptions supporting calculations.

Ingredients

- treat: the expected proportion for the treatment group
- control: the expected proportion for the control group
- power: the required study power
- r: the number in the treatment group divided by the number in the control group.
- sided.test: Use a two-sided test if you wish to evaluate whether the outcome proportion in the treatment group is greater than or less than the outcome proportion in the control group.
- conf.level: defining the level of confidence in the computed result.

Practical exercise

- Clinical question: Among ICU patients with septic shock and AKI without urgent need for dialysis, does an early, compared to delayed, initiation of renal replacement therapy reduce all-cause mortality at 90 days?

ingredients

- treat = .4
- control = .5
- power = .8
- r = 1
- sided.test = 2
- conf.level = .95

Calculation

- The estimated sample size is 192
- Treatment arm: 96
- Control arm: 96

Random allocation

Simple random allocation

- Unequal groups

Random allocation rule

- Equal groups at the end of full recruitment

Block randomization

- Equal groups at any point of recruitment

Learning from mistakes

Comments: RCT

1. Selection bias
2. Serious contradiction in retinopathy inclusion criterion
3. Only one HbA1c measurement at the end of follow up
4. Intervention was not standardized
5. Composite outcome problem
6. Follow-up Duration

Discussion and ideas